Quadratic formula

1 Is the following statement true or false?
'Any quadratic equation that can be solved by completing the square can also be solved by the quadratic formula.'

2 The standard form of a quadratic equation is $ax^2 + bx + c = 0$. Find the values of a, b and c in the quadratic equations below:

a $x^2 + 7x + 10 = 0$

b $4x^2 + 3x = 5$

c $3x^2 - 8x + 2 = 9x - 7$

3 Solve the following equations using the quadratic formula:

a $x^2 + 11x + 28 = 0$

b $x^2 - 5x + 6 = 0$

c $4x^2 - 7x - 15 = 0$

d $x^2 + 5x + \dfrac{9}{4} = 0$

e $-6 - 13x + 5x^2 = 0$

f $x^2 + 8x + 16 = 0$

g $x^2 + 12x + 36 = 0$

h $x^2 - 4x + 4 = 0$

i $x^2 - 10x + 25 = 0$

j $x^2 + 7x + 13 = 0$

k $-x^2 - 3x - 5 = 0$

l $x^2 - 4x + 5 = 0$

m $-x^2 - 5x - 8 = 0$

n $4x^2 + 20x + 25 = 0$

o $4x^2 - 28x + 49 = 0$

p $16x^2 - 24x + 9 = 0$

q $64x^2 + 16x + 1 = 0$

r $2x^2 - 6x + 5 = 0$

s $-2x^2 + 3x - 2 = 0$

t $3x^2 + 7x + 7 = 0$

u $5x^2 - 2x + 1 = 0$

4 Solve the following equations using the quadratic formula. Leave your answers in surd form.

a $x^2 - 5x - 2 = 0$

b $4x^2 - x - 10 = 0$

c $-2x^2 - 15x - 4 = 0$

5 Solve the following equations using the quadratic formula. Round your answers to 1 decimal place.

a $1.8x^2 + 5.2x - 2.3 = 0$

b $x^2 + 7x - 3 = 0$

c $3x\left(x+4\right) = -3x + 4$

6 Consider the equation $x\left(x+9\right) = -20$.

a Solve it by the method of factorisation.

7 Using the quadratic formula, the solutions to a quadratic equation of the form $ax^2 + bx + c = 0$ are given by:

$$x = \frac{-5 \pm \sqrt{5^2 - 4 \times (-7) \times 10}}{2 \times (-7)}$$

a Find the values of a, b and c.

b Write down the quadratic equation that has these solutions.

Applications

8 An object is launched from a height of 90 feet with an initial velocity of 131 feet per second. After x seconds, its height (in feet) is given by

$$h = -16x^2 + 131x + 90$$

Solve for the number of seconds, x, after which the object is 20 feet above the ground. Give your answer to the nearest tenth of a second.